

A Practical Knowledge Graph and LLM-Based Framework for Smart Agromet Advisory Systems

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Abstract

This report presents a practical and implementable framework for building an AI-driven smart agriculture advisory system using Knowledge Graphs and Large Language Models. The work focuses on feasibility, modular implementation, and real-world deployment constraints. Unlike purely theoretical studies, this report outlines step-by-step system design, datasets, tools, and evaluation strategies that can be implemented within a semester timeline.

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1 Introduction

Agriculture depends heavily on environmental conditions such as weather, soil properties, and crop management practices. Modern agrometeorological advisory systems aim to assist farmers by providing actionable recommendations. However, existing systems suffer from fragmented data, lack of semantic reasoning, and poor adaptability.

This report proposes a hybrid system integrating Knowledge Graphs (KGs) and Large Language Models (LLMs) to build an intelligent advisory platform.

2 Problem Statement

Current systems face the following issues:

- Lack of integration between weather, crop, and soil data
- Limited reasoning capabilities
- Static rule-based advisory systems
- Poor natural language interaction

3 Objectives

- Build a domain-specific agricultural knowledge graph
- Integrate real-time weather data
- Use LLMs for natural language advisory generation
- Implement a retrieval-based system to reduce hallucination

4 System Overview

The system consists of four main components:

1. Data Collection
2. Knowledge Graph Construction
3. Retrieval-Augmented Generation
4. User Interface

5 Dataset Selection

Feasible datasets include:

- FAO agricultural datasets
- IMD weather data
- Kaggle crop datasets

6 Knowledge Graph Design

Entities:

- Crop
- Soil
- Weather
- Advisory

Relations:

- affects
- requires
- causes

7 Implementation Tools

- Python
- Neo4j or RDF store
- HuggingFace Transformers
- FAISS

8 Relation Extraction

We use a BERT-based model for extracting relationships from agricultural text.

9 RAG Pipeline

Steps:

1. Query input
2. Retrieval
3. LLM generation

10 System Workflow

1. Data ingestion
2. KG update
3. Query processing
4. Advisory generation

11 Evaluation Metrics

- Accuracy
- Latency
- Precision/Recall

12 Implementation Plan

Week-wise plan:

- Week 1-2: Dataset collection
- Week 3-4: KG construction
- Week 5-6: Model integration
- Week 7-8: Evaluation

13 Results (Expected)

Improved advisory accuracy and better interpretability.

14 Conclusion

This system provides a scalable and practical solution for smart agriculture advisory.